LAB 4 Simulated Internet (10.10.0.0/16) & Private LANs with Stateful vs Stateless Services

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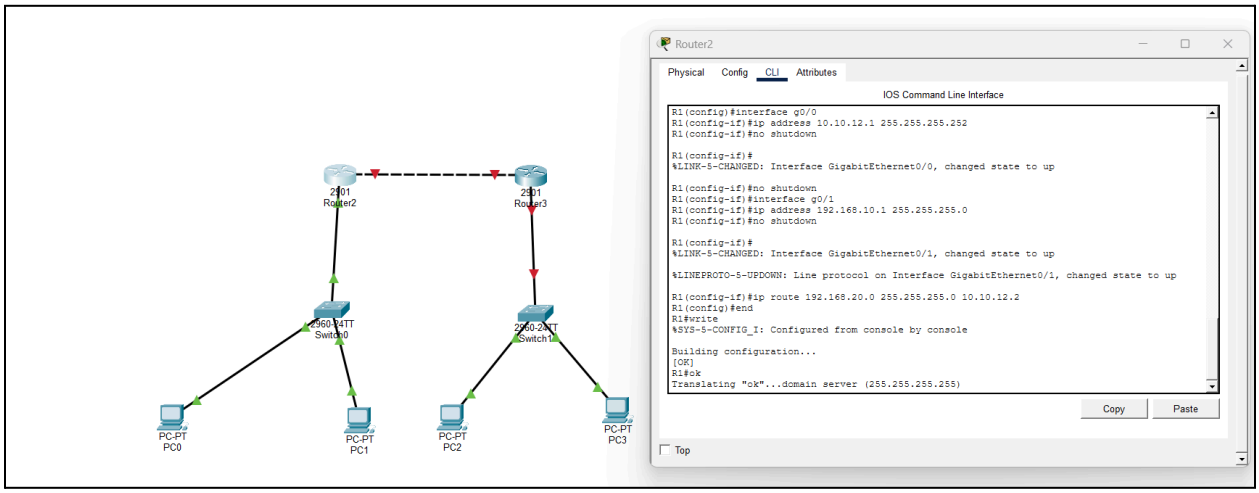
🔧 PART 1 — CISCO-STYLE NETWORK CONFIGURATION

We will configure static routing + NAT to allow cross-site reachability.

🧱 R1 CONFIGURATION (LAN A)

```
None
enable
conf t

hostname R1

interface g0/0
 ip address 10.10.12.1 255.255.255.252
 ip nat outside
 no shutdown
```

```
interface g0/1
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 no shutdown

ip route 192.168.20.0 255.255.255.0 10.10.12.2

access-list 10 permit 192.168.10.0 0.0.0.255
ip nat inside source list 10 interface g0/0 overload

end
write memory
```

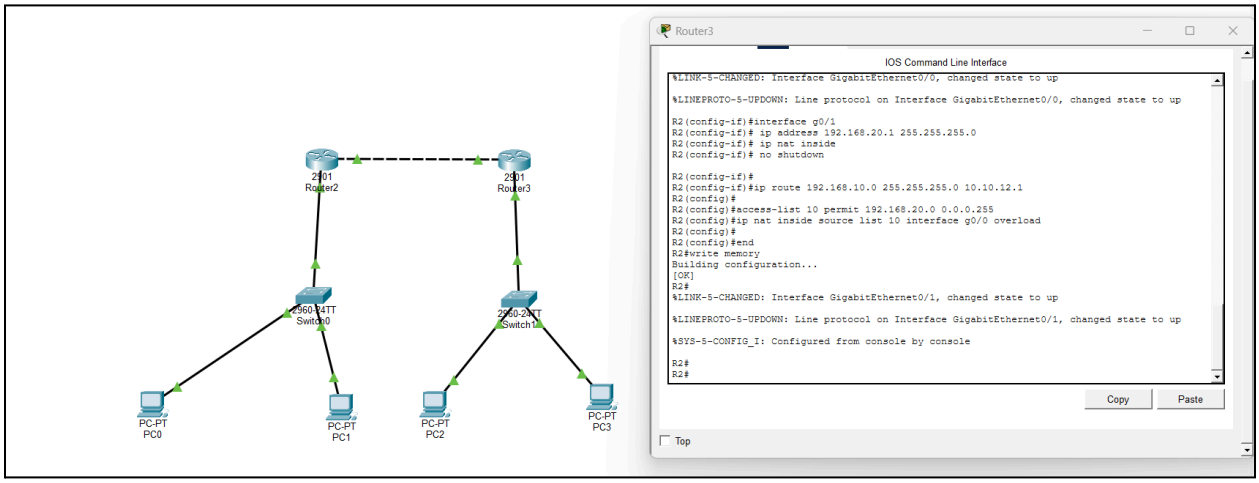


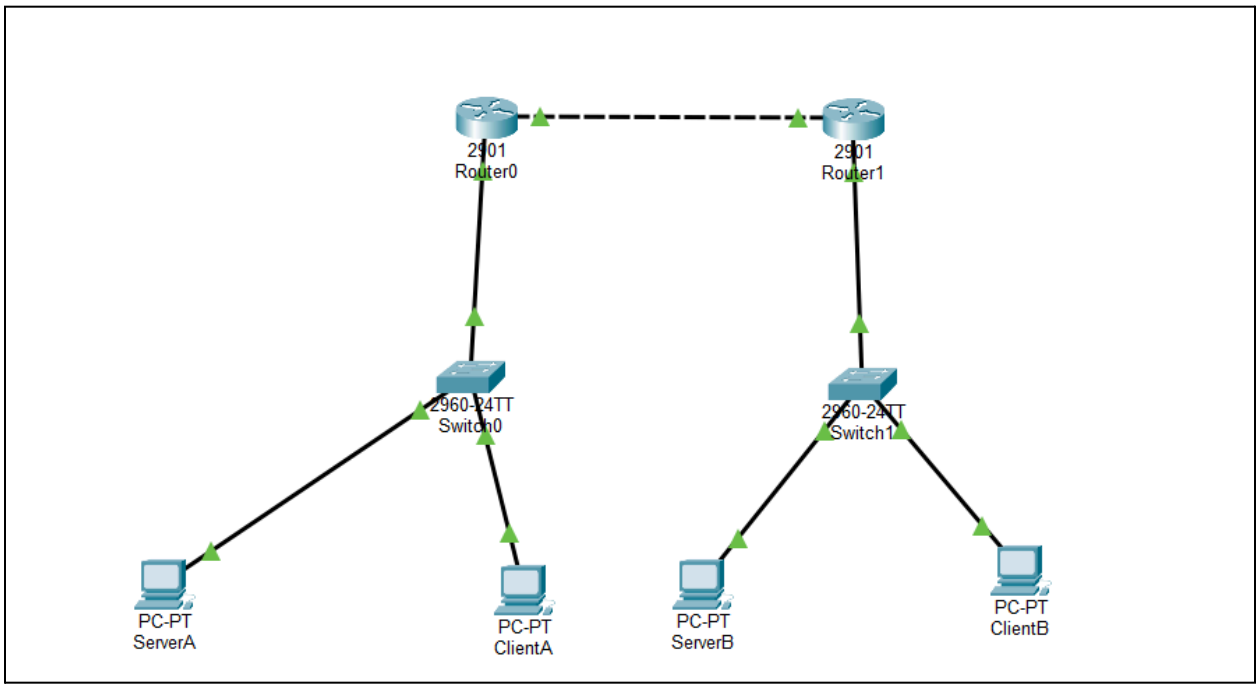
🧱 R2 CONFIGURATION (LAN B)

None

```
enable
conf t

hostname R2

interface g0/0
 ip address 10.10.12.2 255.255.255.252
 ip nat outside
 no shutdown

interface g0/1
```

```
 ip address 192.168.20.1 255.255.255.0
 ip nat inside
 no shutdown

ip route 192.168.10.0 255.255.255.0 10.10.12.1

access-list 10 permit 192.168.20.0 0.0.0.255
ip nat inside source list 10 interface g0/0 overload

end
write memory
```





🧪 ROUTING & NAT VERIFICATION

On R1 & R2:

```
None
show ip route
show ip nat translations
```

R1

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.12.0/30 is directly connected, GigabitEthernet0/0
L       10.10.12.1/32 is directly connected, GigabitEthernet0/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/1
L       192.168.10.1/32 is directly connected, GigabitEthernet0/1
S    192.168.20.0/24 [1/0] via 10.10.12.2

R1#show ip nat translations
```

R2

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.12.0/30 is directly connected, GigabitEthernet0/0
L       10.10.12.2/32 is directly connected, GigabitEthernet0/0
S    192.168.10.0/24 [1/0] via 10.10.12.1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1

R2#show ip nat translations
```

From ClientA:

```
None
ping 192.168.20.10  ← Remote server reachable
```

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 10.10.12.2: bytes=32 time<1ms TTL=126
Reply from 10.10.12.2: bytes=32 time<1ms TTL=126
Reply from 10.10.12.2: bytes=32 time<1ms TTL=126
Reply from 10.10.12.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

💻 PART 2 — IaC + MOCKUP-INFRA DEPLOYMENT (mockup-infra/README.md)

We will now set up mockup infra using the paths you specified:

```
None
automation/mockup-infra
automation/week02-stateless-stateful/phase1-mockup
```

🛠️ CLONE REPO

```shell
git clone https://github.com/boonsup/automation.git
cd automation
```

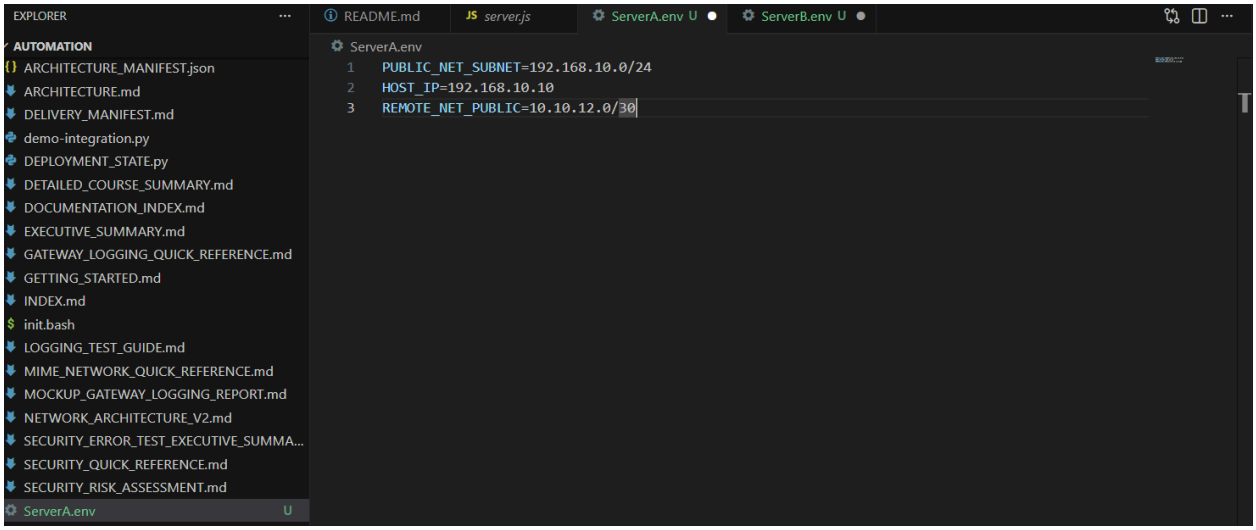
🧰 VS CODE — OPEN PROJECT

In VS Code, open the `automation` folder.

📄 CREATE SERVER-SPECIFIC ENV FILES

ServerA (`ServerA.env`)

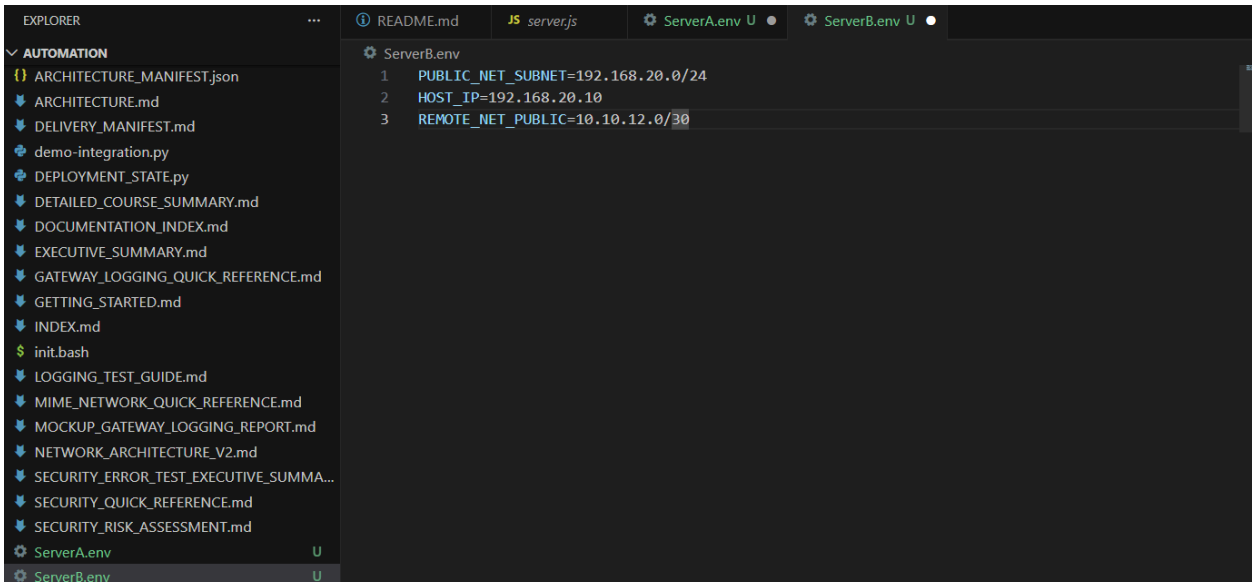
```
None
# Private LAN A
PUBLIC_NET_SUBNET=192.168.10.0/24
HOST_IP=192.168.10.10
REMOTE_NET_PUBLIC=10.10.12.0/30
```



ServerB (`ServerB.env`)

None

```
# Private LAN B
PUBLIC_NET_SUBNET=192.168.20.0/24
HOST_IP=192.168.20.10
REMOTE_NET_PUBLIC=10.10.12.0/30
```



🚀 DEPLOY MOCKUP-INFRA

PowerShell (Windows Admin)

```
None
cd mockup-infra
python manage.py init
python manage.py deploy --profile week02-stateless-stateful
```

```
(.venv) PS C:\67338277-1\automation\mockup-infra> python manage.py init

🔧 Initializing Mockup Infrastructure...
==================================================

🔐 [Layer 5/6 - Session] Generating TLS Certificates...
C:\67338277-1\automation\mockup-infra\manage.py:139: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version
. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  .not_valid_before(datetime.utcnow())
C:\67338277-1\automation\mockup-infra\manage.py:140: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version
. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  .not_valid_after(datetime.utcnow() + timedelta(days=365))
✅ Certificates generated successfully:
   - Private Key: C:\67338277-1\automation\mockup-infra\certs\server.key
   - Certificate: C:\67338277-1\automation\mockup-infra\certs\server.crt
   - Common Name: api.mockup.test
   - Subject Alt Names: localhost, api.mockup.test, 127.0.0.1, 172.18.0.1, 172.19.0.1

🌐 [Layer 3 - Network] Testing Isolation...
🔧 Executing: podman network inspect private_net
[
    {
        "name": "private_net",
        "id": "522abcf8eb72b42e0586d67ea72a1c8b91e22c948dbdaf6cacf217323c58731d",
        "driver": "bridge",
        "network_interface": "podman4",
        "created": "2026-03-11T15:45:42.146850333+07:00",
        "subnets": [
            {
                "subnet": "10.89.1.0/24",
                "gateway": "10.89.1.1"
            }
        ],
        "ipv6_enabled": false,
        "internal": true,
        "dns_enabled": true,
        "ipam_options": {
            "driver": "host-local"
        },
        "containers": {}
    }
]
✅ Private network is isolated (internal: true)
```

```
🚀 Deploying Mockup Infrastructure Stack
   Bare Metal → L3 → L4 → L5/6 → L7
==================================================

🌐 [Layer 3 - Network] Testing Isolation...
🔧 Executing: podman network inspect private_net 2>/dev/null || echo '{}'
The system cannot find the path specified.
'{}'
⚠️ Private network not found or not isolated. Creating networks...
🔧 Executing: podman network create public_net 2>/dev/null || true
The system cannot find the path specified.
'true' is not recognized as an internal or external command,
operable program or batch file.
🔧 Executing: podman network create --internal private_net 2>/dev/null || true
The system cannot find the path specified.
'true' is not recognized as an internal or external command,
operable program or batch file.
✅ Networks created successfully
🔧 Executing: podman-compose up -d --remove-orphans
```

Bash (Linux)

```shell
cd mockup-infra
chmod +x scripts/*.sh
./scripts/init-env.sh
python manage.py deploy --profile week02-stateless-stateful
```

🟡 VERIFY CONTAINERS

```none
podman ps
```

Should show:

Container	Port
stateless-api	3000
stateful-api	3001

```
(.venv) PS C:\6733827-1\automation\mockup-infra> podman ps
CONTAINER ID  IMAGE                                       COMMAND               CREATED        STATUS         PORTS                                       NAME
S
c1e8a9518792  localhost/mockup-infra_public_app:latest    python app.py         14 seconds ago  Up 11 seconds  80/tcp                                      mock
up-public-web
150f3a91f0c6  localhost/mockup-infra_intranet_api:latest  python api.py         13 seconds ago  Up 10 seconds  5000/tcp                                    mock
up-intranet-api
3a19d92ebad9  docker.io/library/nginx:alpine              nginx -g daemon o...  12 seconds ago  Up 9 seconds   0.0.0.0:443->443/tcp, 0.0.0.0:8080->80/tcp  mock
up-gateway
f40f7f37c650  localhost/stateless-api:v2                  node start-statel...  11 seconds ago  Up 9 seconds   3000/tcp                                    stat
eless-api
3d182d91604d  localhost/stateful-api:v2.1                 node start-statef...  10 seconds ago  Up 8 seconds   3001/tcp                                    stat
eful-api
```

🧪 PART 3 — TESTING STATELESS vs STATEFUL (week03-stateless-stateful/phase1-mockup/README.md)

⚫ LOCAL TESTS (Private LAN)

From ClientA:

```shell
curl http://192.168.10.10:3000/info
```

```
StatusCode        : 200
StatusDescription : OK
Content           : {"status":"success","message":"Stateless server
                    information","data":{"server":"Stateless Mock Server v1.0","timestamp":
                    "2026-03-11T10:34:06.308Z","requestCount":1,"clientId":"anonymous","ran
                    domValue":...
RawContent        : HTTP/1.1 200 OK
                    Content-Security-Policy: default-src 'self';base-uri 'self';font-src
                    'self' https: data:;form-action 'self';frame-ancestors 'self';img-src
                    'self' data:;object-src 'none';script-src 's...
Forms             : {}
Headers           : {[Content-Security-Policy, default-src 'self';base-uri 'self';font-src
                    'self' https: data:;form-action 'self';frame-ancestors 'self';img-src
                    'self' data:;object-src 'none';script-src 'self';script-src-attr
                    'none';style-src 'self' https:
                    'unsafe-inline';upgrade-insecure-requests],
                    [Cross-Origin-Opener-Policy, same-origin],
                    [Cross-Origin-Resource-Policy, same-origin], [Origin-Agent-Cluster,
                    ?1]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 422
```

→ Stateless

From ClientA:

```shell
curl -X POST http://192.168.10.10:3001/session
```

```
HP on D:\..\..\..\..\phase1-mockup main
# curl.exe -X POST http://localhost:3001/session
{"status":"error","message":"Validation failed","errors":[{"field":"body","message":"Missing
 required fields: userId"}],"timestamp":"2026-03-11T10:36:15.641Z"}
HP on D:\..\..\..\..\phase1-mockup main
```

Store the returned `Session-ID`, then:

```shell
curl http://192.168.10.10:3001/data -H "Session-ID: <id>"
```

🔵 REMOTE TESTS (Via Simulated Internet)

From ClientA:

```
None
curl http://10.10.12.2:3000/info
```

```
# curl.exe http://localhost:3000/info
{"status":"success","message":"Stateless server information","data":{"server":"Stateless Moc
k Server v1.0","timestamp":"2026-03-11T10:39:52.581Z","requestCount":2,"clientId":"anonymous
","randomValue":0.18539100360907024,"message":"I have no memory of previous requests. Each r
equest is independent.","note":"This is a stateless server - I don't remember you from previ
ous requests."},"timestamp":"2026-03-11T10:39:52.581Z"}
HP on D:\..\..\..\..\phase1-mockup main
#
```

That will target **ServerB stateless** — NAT + routing should deliver

---

Session Test:

```
None
curl -X POST http://10.10.12.2:3001/session
```

```
HP on D:\..\..\..\..\phase1-mockup main
# curl.exe -X POST http://localhost:3001/session -H "Content-Type: application/json" -d "{`"
userId`": `"student1`"}"
{"status":"error","message":"Internal server error","errors":[],"timestamp":"2026-03-11T10:4
1:03.388Z"}
HP on D:\..\..\..\..\phase1-mockup main
#
```

→ ServerB returns a session ID

Then:

```
None
curl http://10.10.12.2:3001/data -H "Session-ID: <id>"
```

Check session persists across NAT

⚠️ TROUBLESHOOTING

❗ Cannot Reach Remote

- Gateway misconfiguration
- NAT ACL incorrect
- Wrong default gateway on clients

Check:

```
None
show ip route
ping 10.10.12.2 from R1
```

❗ Sessions Lose State

- Missing header
- Wrong container port
- Firewall blocking port 3001

🧠 ENGINEERING REFLECTION

1. **Stateless services don't depend on prior data**
2. **Stateful services require session headers**
3. NAT shouldn't affect logical session identity

📄 LAB README.md (for repository)

```
None
# Lab 4 — Stateful vs Stateless Across Public & Private Subnets

## Architecture
Two LANs connected via a simulated internet (10.10.12.0/30)
- LAN A: 192.168.10.0/24
- LAN B: 192.168.20.0/24
Routers provide static routing + NAT.

## Steps

### 1. Setup Network
Configure R1 and R2:
- Static routes
- NAT overload
- Verify ping

### 2. Deploy IaC
Configure `ServerA.env` and `ServerB.env`
Deploy via VS Code, PowerShell or Bash using:
```

python manage.py deploy --profile week02-stateless-stateful

```
None

### 3. Validate
- Local stateless vs stateful
- Remote stateless vs stateful
- Test session persistence
- Capture NAT translations

## Troubleshooting
- Check interface status, NAT ACLs
- Verify client gateways

## Extensions
- OSPF instead of static routing
- Load balancer in front of APIs
```